

PH-1843

B. Tech. (First Semester)

EXAMINATION, 2025-26

ENGG. PHYSICS-I

Time : Two Hours

Maximum Marks : 50

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *five* questions in maximum **30** words. Each question carries 4 marks. $5 \times 4 = 20$

1. A particle of rest mass m_0 moves with speed $\frac{c}{\sqrt{2}}$. Calculate its mass, momentum, total energy and kinetic energy.

P. T. O.

2. Define dispersive power of grating.
3. What do you mean by Nicol's prism ?
4. Write the expression for space and time by Lorentz transformation.
5. What are the main uses of lasers ?
6. A man weighs 50 kg on the earth, when he is in a rocket ship in flight his mass is 50.5 kg measured by an observer on earth. What is the speed of the rocket ?
7. What do you mean by polarisation of light ?
8. Give the types of optical fiber.
9. Give the significance of null result from Michelson-Morley experiment.
10. Write the difference between interference and diffraction.

Note

1.

2.

3.

4.

Section—B**(Short Answer Type Questions)**

Note : Attempt any *two* questions in maximum **150** words. Each question carries 7.5 marks. $2 \times 7.5 = 15$

1. Obtain the time dilation formula of special theory of relativity by Lorentz transformation.
2. Define construction and reconstruction image of Holography.
3. Define Einstein's A and B coefficients.
4. What is interference of light ? Derive reflected light to interference on the basis of wave theory of light ?

Section—C**(Long Answer Type Questions)**

Note : Attempt any *one* question in maximum **300** words. Each question carries 15 marks. $1 \times 15 = 15$

1. Deduce the law of relativistic velocity addition theorem.

P. T. O.

2. Explain how Newton's ring method can be used for determining the wavelength of monochromatic light.
3. Define diffraction of light. Drive the single slit diffraction.
4. Describe the principle and working of Ruby laser. Explain why a helium neon laser is superior than Ruby laser.

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B. Tech. (First Semester)

EXAMINATION, 2024-25

ENGINEERING PHYSICS-I

Time : Two Hours

Maximum Marks : 50

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *five* questions. Each question carries 4 marks. $5 \times 4 = 20$

1. What are the coherent sources of light ?

P. T. O.

2. Define population inversion.
3. Give the *two* postulates of Einstein's special theory of relativity.
4. What is holography ?
5. Explain the resolving power of an optical instrument.
6. Explain interference of light.
7. Explain the term Lorentz-Fitzgerald (length) contraction.
8. What is the unique characteristics of a Laser ?
9. What is the difference between Fresnel and Fraunhofer's diffraction ?
10. Calculate the momentum of photon of energy 5 eV.

Section—B

(Short Answer Type Questions)

Note : Attempt any two questions. Each question carries $7\frac{1}{2}$ marks. $2 \times 7\frac{1}{2} = 15$

1. ✓ Show that the clock moving velocity v relative to an observer appears as slow by a

factor of $\sqrt{\frac{1-v^2}{c^2}}$.

2. ✓ Write the application of optical fibre communication.

3. ✓ Explain spontaneous and stimulated emission of radiation.

4. Describe with necessary theory the Fresnel type of diffraction due to a straight edge.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. Prove that $x^2 + y^2 + z^2 - c^2 t^2$ is invariant under Lorentz transformation.
2. Describe the construction and working of Michelson's interferometer and explain the negative result of Michelson's interferometer.
3. Define optical rotation. Give an outline of Fresnel's theory of optical rotation.
4. Explain Einstein's A and B coefficients and discuss their physical significance. Derive a relation between these coefficients.

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PH-101/1843

**B. Tech. (First Semester)
EXAMINATION, 2019
ENGINEERING PHYSICS**

Time : Three Hours

Maximum Marks : 50

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any *ten* questions. Each question carries 2 marks. $10 \times 2 = 20$

1. What is mass-energy equivalence ?
2. Explain the basic postulates of special theory of relativity.
3. Show that in the phenomenon of interference, the law of conservation of energy holds.

4. Distinguish between the Fresnel and Fraunhofer class diffraction.
5. How is a concave reflection grating superior over a plane diffraction grating ?
6. A rod of length 10 m long is moving along its length with a velocity of $0.6C$. Calculate its length as it appears to an observer at rest.
7. Two coherent sources whose intensity ratio is $36 : 1$ produce interference fringes. Deduce the ratio of maximum intensity to minimum intensity.
8. What do you mean by double refraction ?
9. What do you mean by population inversion ?
10. Explain the principle of holography.
11. What is the basic principle of fibre optics ?
12. Define numerical aperture.
13. What are the properties of laser radiation ?
14. What are Newton's rings ?
15. Discuss Lorentz transformations.

and

Section—B

(Long Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 15 marks. $2 \times 15 = 30$

1. Describe Michelson-Morley experiment and Physical significance of its results.
2. Describe Fresnel's Bi-prism with a neat diagram and explain how wavelength of monochromatic light is determined by it.
3. Discuss Einstein's co-efficients. Derive a relation between them.
4. (a) How does one construct and reconstruct a hologram ?
(b) Explain acceptance angle and acceptance cone of a filter.

PH-101/1843

B. Tech. (Semester-I) Examination-2018

Engineering Physics

Time: Three Hours

Maximum Marks: 50

Note: Attempt questions from all the sections.

Section-A

(Short Answer Type Questions)

Note: Attempt any ten questions. Each question carries 2 marks. (2x10=20)

1. Define the length contraction. (2013)

2. Calculate the rest mass, relativistic mass and momentum of a photons of energy 5ev.

3. Explain the relativistic addition of velocities.

4. What is interference of light?
5. Explain Rayleigh's criterion of resolution. (2015, 2014, 2013)
6. Describe the construction and action of the ruby laser?
7. What is Nicol prism?
8. Deduce an expression for the variation of mass with velocity. (2013)
9. Distinguish between ordinary light and laser radiation.
10. How many lines per centimetre are there in a plane transmission grating which gives 1st order of light of wave length $6000\text{\AA}$ at angle of diffraction 30° ?
11. What is Hologram? Explain with diagram the basic principle of holography.

Explain the colours when a thin film illuminated by white light is observed in reflected light. (2015)

Discuss briefly Michelson-Morely experiment and mention its outcome. (2014, 2012)

The rest mass of a proton is $1.67 \times 10^{-27} \text{ kg}$. At what speed will its mass be double its rest mass?

Discuss briefly the working principle of a polarimeter.

Section-B

(Long Answer Type Questions)

Note: Attempt any two questions. Each question carries 15 marks. (15x2=30)

Prove the relation $E^2 - P^2 C^2 = M^2 C^4$ where P is the momentum. (2015, 2012)

Explain phenomenon of diffraction and distinguish Fresnel and Fraunhofer diffractions. Obtain intensities of diffraction pattern in fraunhofer diffraction due to a single slit.

3. Describe and explain the phenomenon of optical rotation. Show that $\theta = \frac{\pi l}{\lambda}(\mu_L - \mu_R)$, where symbols have their usual meaning.
4. Explain basic principle of optical fibre, Discuss fibre classification.

Section-B

(Long Answer Type Questions)

Note: Attempt any two questions. Each question carries

(15×2=30)

15 marks.

1. Prove the relation $E \cdot P \cdot C = M \cdot C$ where P is the

transmission coefficient.

2. Explain the phenomenon of optical activity and its

relation to the optical activity of substances.

3. Explain the phenomenon of optical activity and its

relation to the optical activity of substances.